

Homework #6

[Your name]

[Total - 100 pts]

Question 1: For each of the following, (i) verify each distribution belongs to the exponential family, (ii) find the MLE for θ , and (iii) find the limiting distribution of the MLE.

(a) Beta distribution: $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Beta}(\theta, 1)$, $\theta > 0$.

(b) Pareto distribution: $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Pareto}(1, \theta)$, $\theta > 2$.

[1.a] *Solution*

■

[1.b] *Solution*

■

Question 2: Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Uniform}(-\theta, \theta)$ where $\theta > 0$. Answer the following:

(a) Show that the statistic $Y = \max\{|X_i|\}$ for $1 \leq i \leq n$ and $\theta \in (0, +\infty)$ is a sufficient statistic.

(b) Denote the order statistics as $X_{(1)} < \dots < X_{(n)}$ for $n \geq 2$, and define an estimator for θ as:

$$\hat{\theta} = c_n \cdot (X_{(n)} - X_{(1)})$$

Find the value of c_n such that the estimator is unbiased for θ (i.e., $\mathbb{E}(\hat{\theta}) = \theta$).

[2.a] *Solution*

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[2.b] *Solution*

■

Question 3: Find a sufficient statistic:

(a) $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(\theta)$, for $\theta > 0$, where the pdf is given by

$$\text{pdf}(x; \theta) = \frac{1}{\theta} e^{-x/\theta} \mathbf{1}\{x \in (0, \infty)\}.$$

(b) $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Gamma}(\alpha, \beta)$, for $\alpha > 0$ and $\beta > 0$, where the pdf is given by

$$\text{pdf}(x; \alpha, \beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta} \mathbf{1}\{x \in (0, \infty)\}.$$

[3.a] *Solution*

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[3.b] *Solution*

■

Question 4: Consider an exponential distribution $\text{Exp}(\mu, \sigma)$ with two parameters, where $-\infty < \mu < +\infty$ and $\sigma > 0$. The marginal pdf is given by

$$\text{pdf}(x; \mu, \sigma) = \frac{1}{\sigma} e^{-(x-\mu)/\sigma} \cdot \mathbf{1}\{x \in (\mu, \infty)\}.$$

Suppose that you are unable to observe the individual random samples, but you can observe their order statistics $X_{(1)} < \cdots < X_{(r)}$ for $1 \leq r < n$.

- (a) Does $\text{Exp}(\mu, \sigma)$ belong to the exponential family? Explain.
- (b) Find a sufficient statistic for (μ, σ) , where $(\mu, \sigma) \in \mathbb{R} \times \mathbb{R}_+$.

[4.a] *Solution*

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[4.b] *Solution*

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